

# FACE: ArcFace White-box Geometric Identity Optimization

## Frozen iResNet-100 identity-distance results with downstream InstructPix2Pix evaluation

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FACE optimizes  $Z = 1 - \text{cosine\_similarity}$  between original and perturbed full-image ArcFace iResNet-100 embeddings. The optimized loss is exactly  $\text{loss} = -Z$ . ArcFace weights are frozen; only geometry parameters are optimized.

No landmarks, face detection, alignment, visual counter-loss, VAE objective, or UNet objective is used. Prompts are retained only for downstream InstructPix2Pix evaluation.

### Run matrix and final values

| face     | prompt               | final Z | best Z | cosine sim | score % | SSIM   | PSNR   | edit SSIM | max disp |
|----------|----------------------|---------|--------|------------|---------|--------|--------|-----------|----------|
| face_002 | add black sunglasses | 0.0242  | 0.0242 | 0.9758     | 97.580  | 0.9786 | 25.512 | 0.6861    | 5.5964   |
| face_002 | add headphones       | 0.0292  | 0.0293 | 0.9708     | 97.077  | 0.9679 | 23.752 | 0.6030    | 7.6190   |
| face_005 | add black sunglasses | 0.0316  | 0.0317 | 0.9684     | 96.836  | 0.9636 | 23.174 | 0.6932    | 13.032   |
| face_005 | add headphones       | 0.0311  | 0.0311 | 0.9689     | 96.892  | 0.9613 | 22.914 | 0.6513    | 14.629   |

## Case image strips

### *face\_002 / add black sunglasses*



### *face\_002 / add headphones*



### *face\_005 / add black sunglasses*

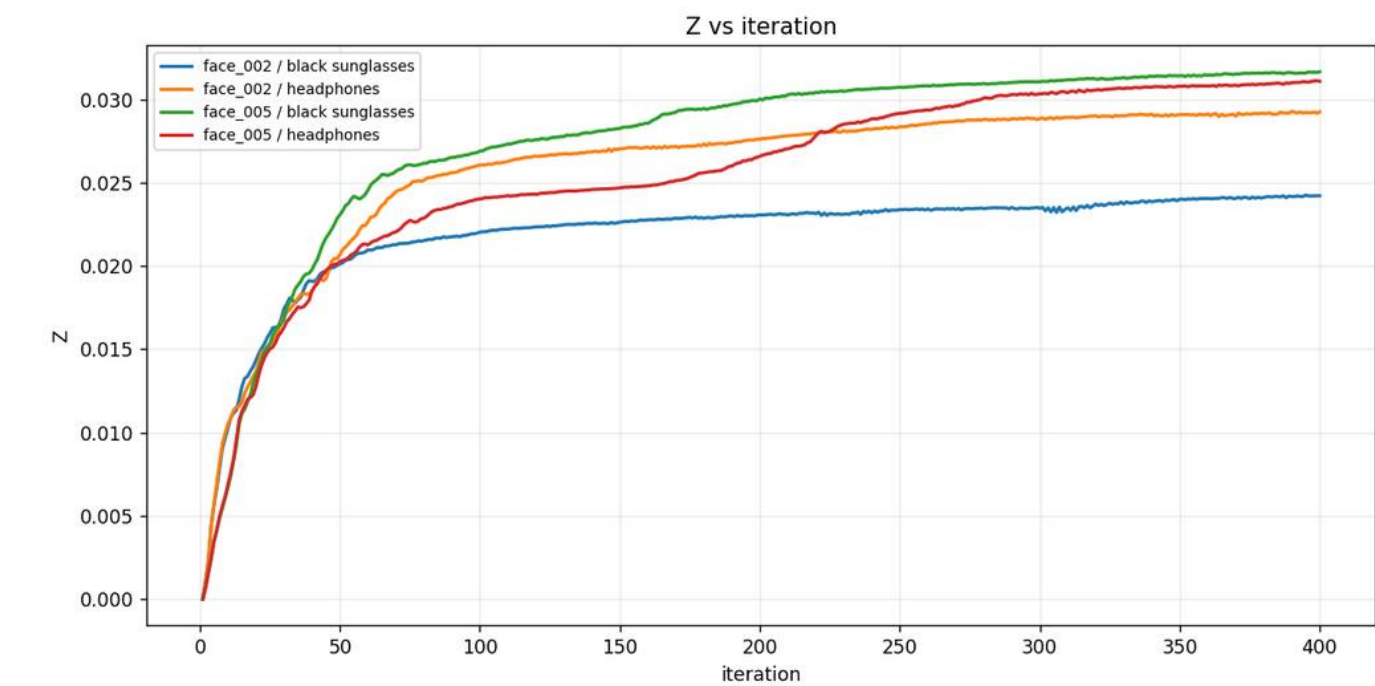


### *face\_005 / add headphones*

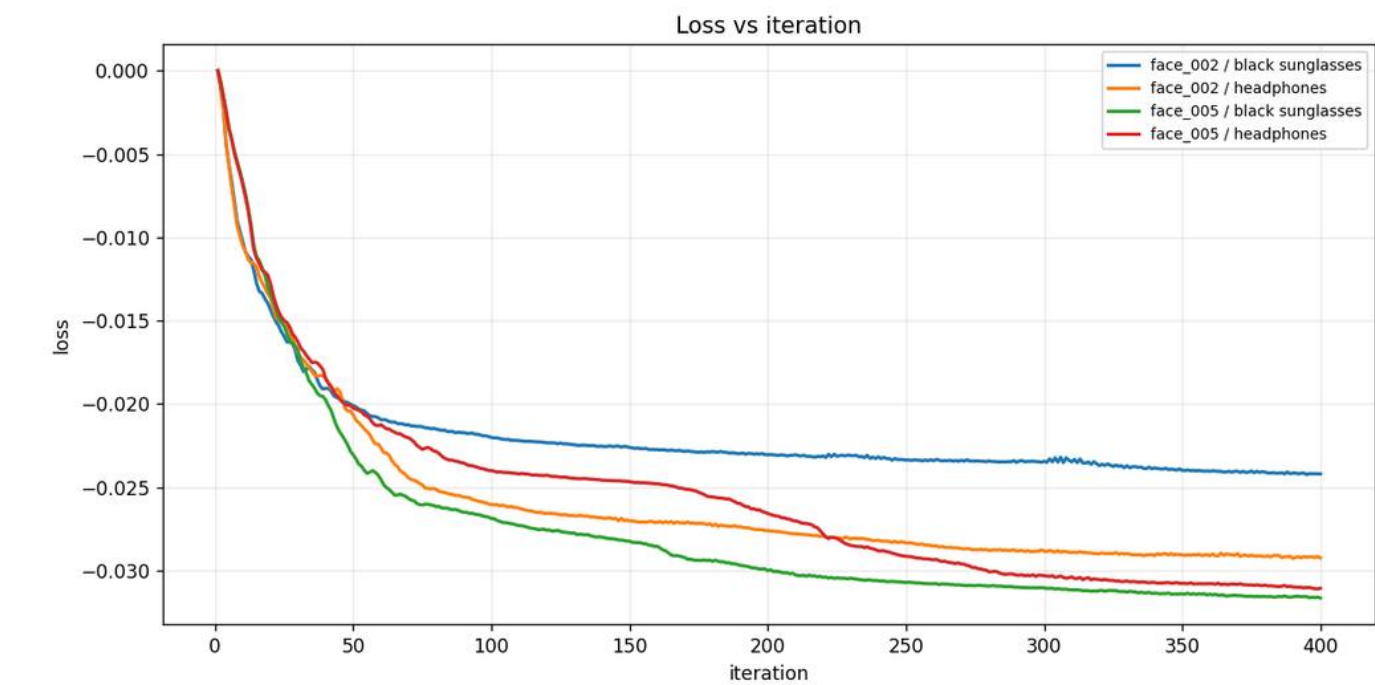


# Graphs

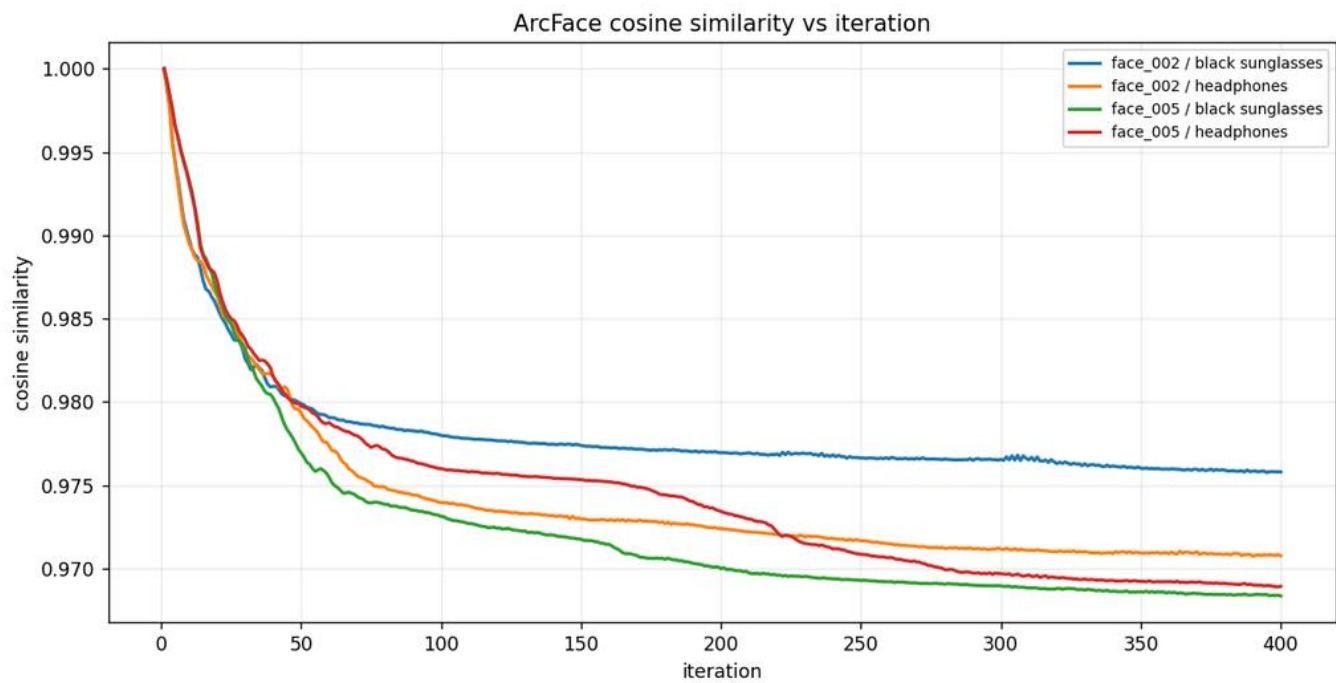
## Z vs iteration



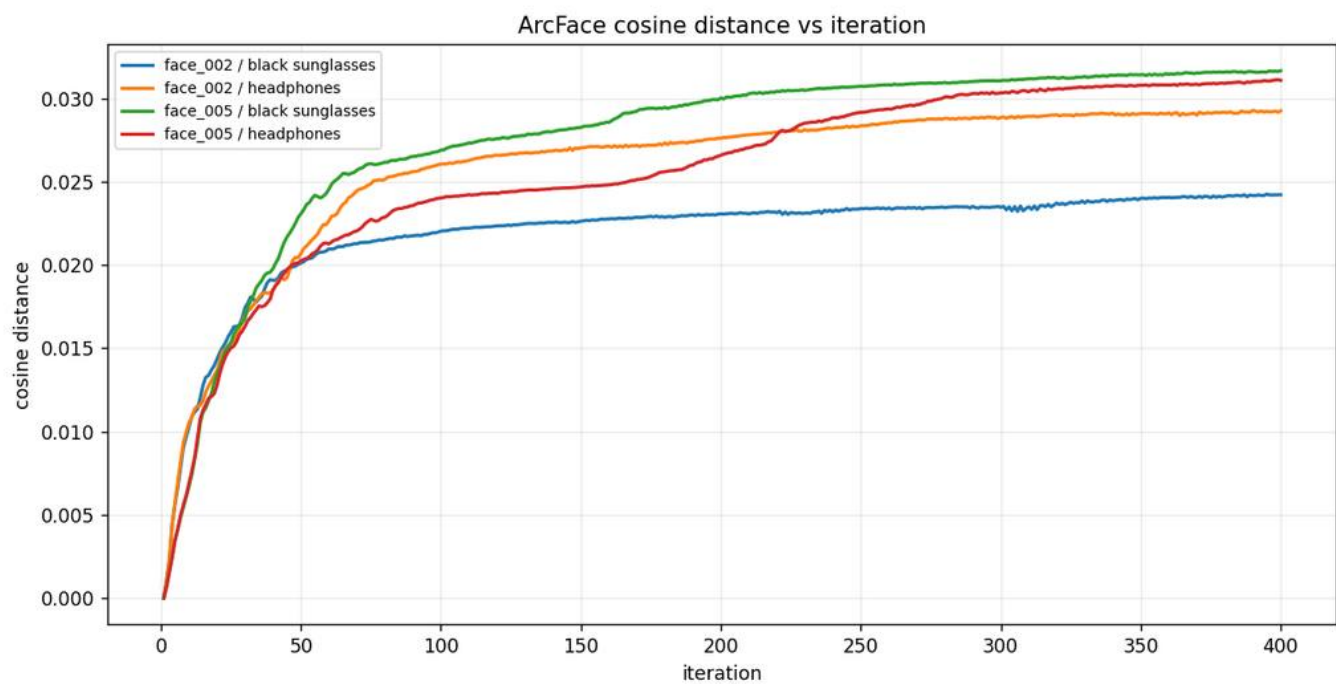
## Loss vs iteration



## ArcFace cosine similarity vs iteration

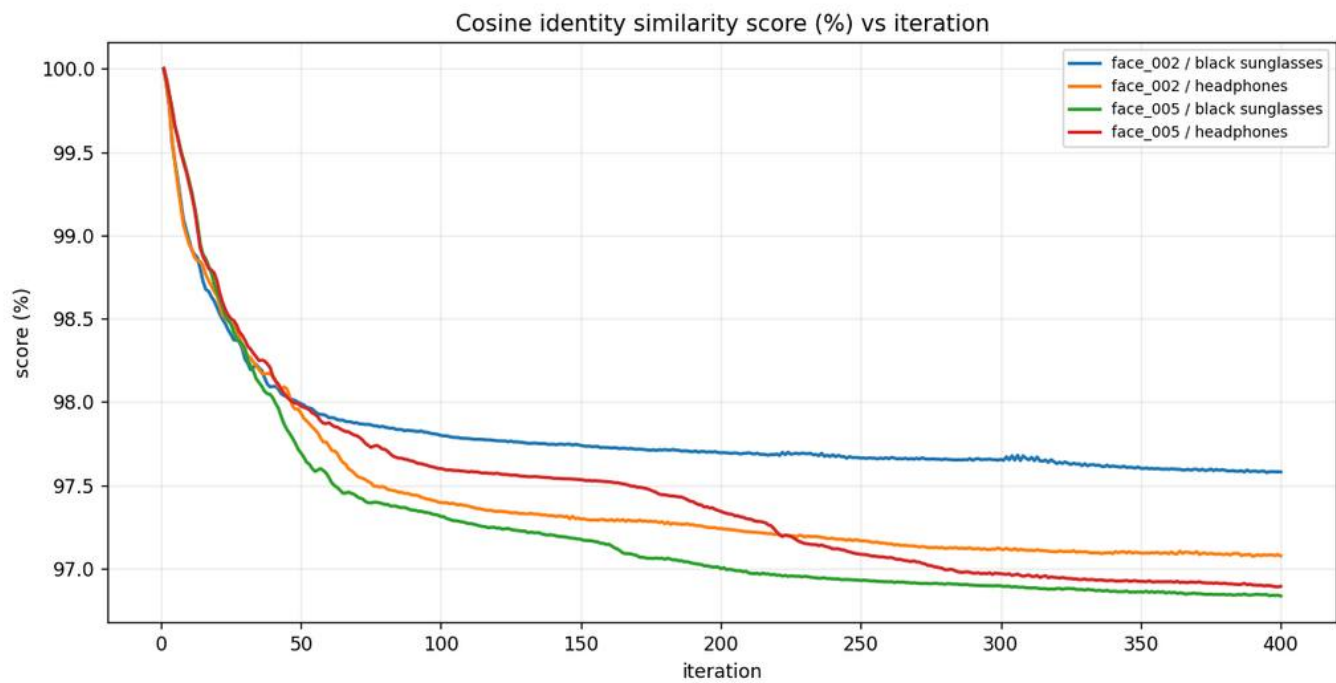


### ***ArcFace cosine distance vs iteration***

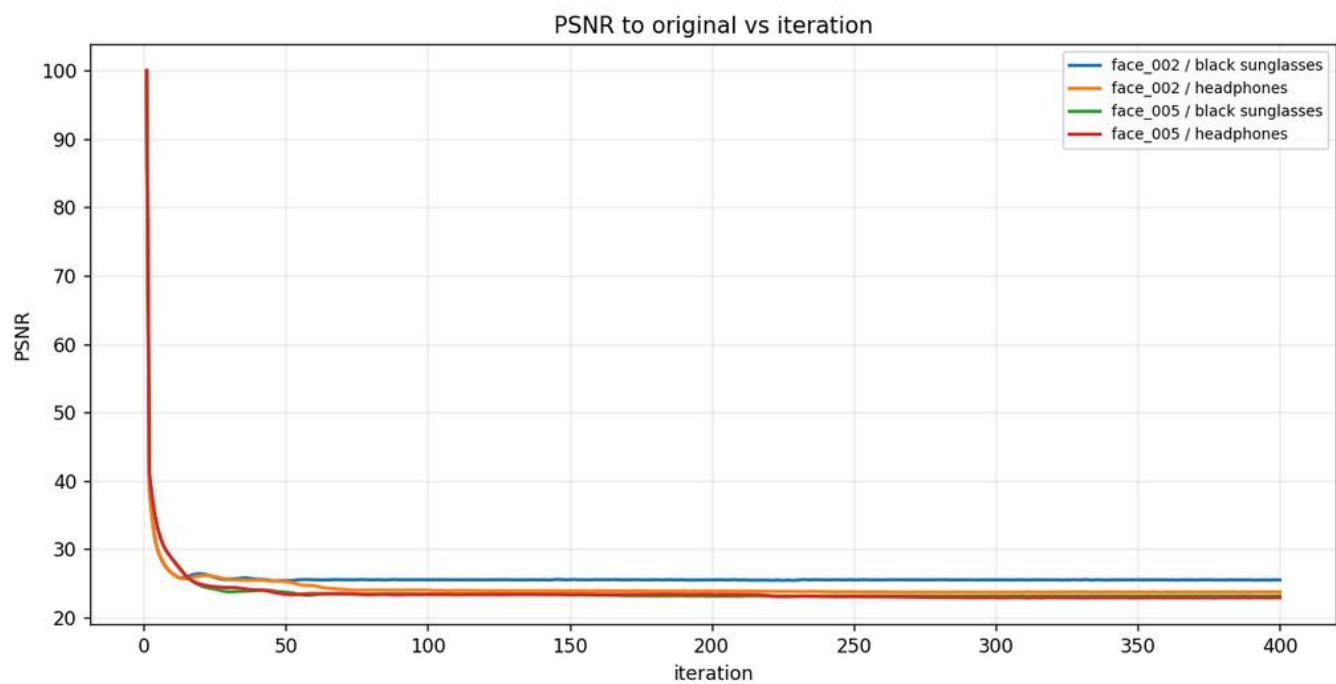


### ***Cosine identity similarity score (%) vs iteration***

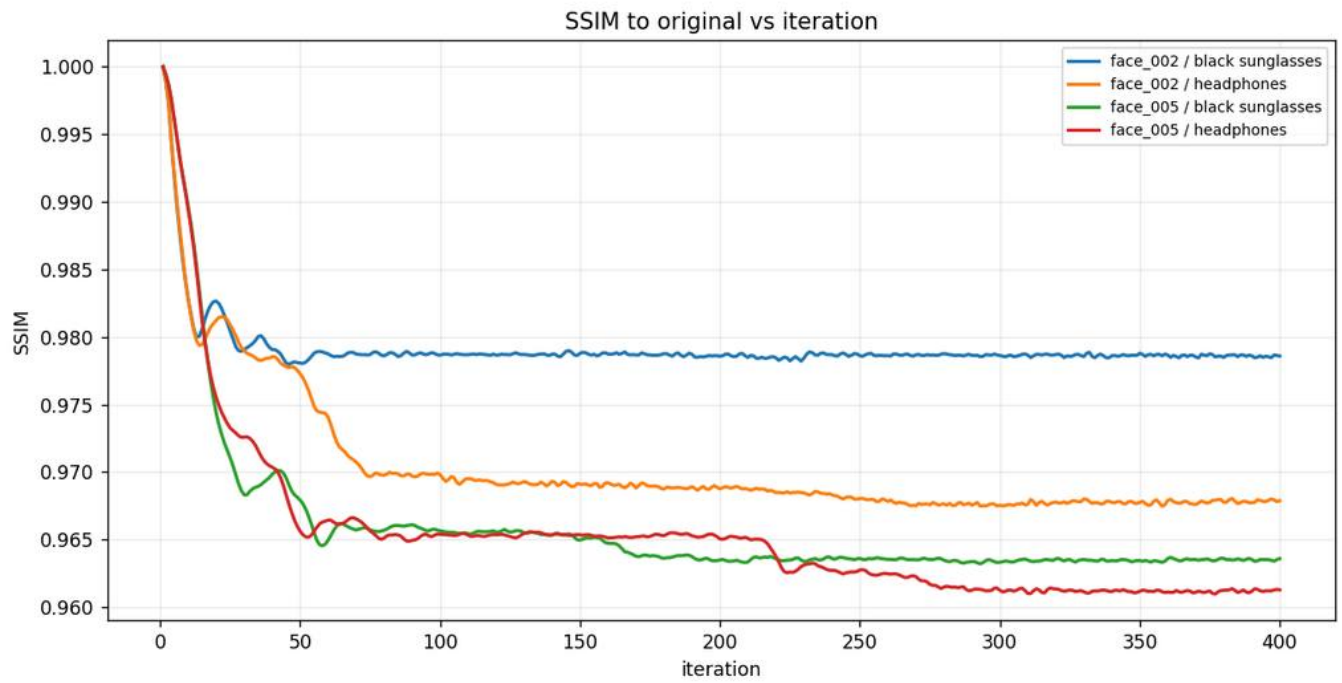




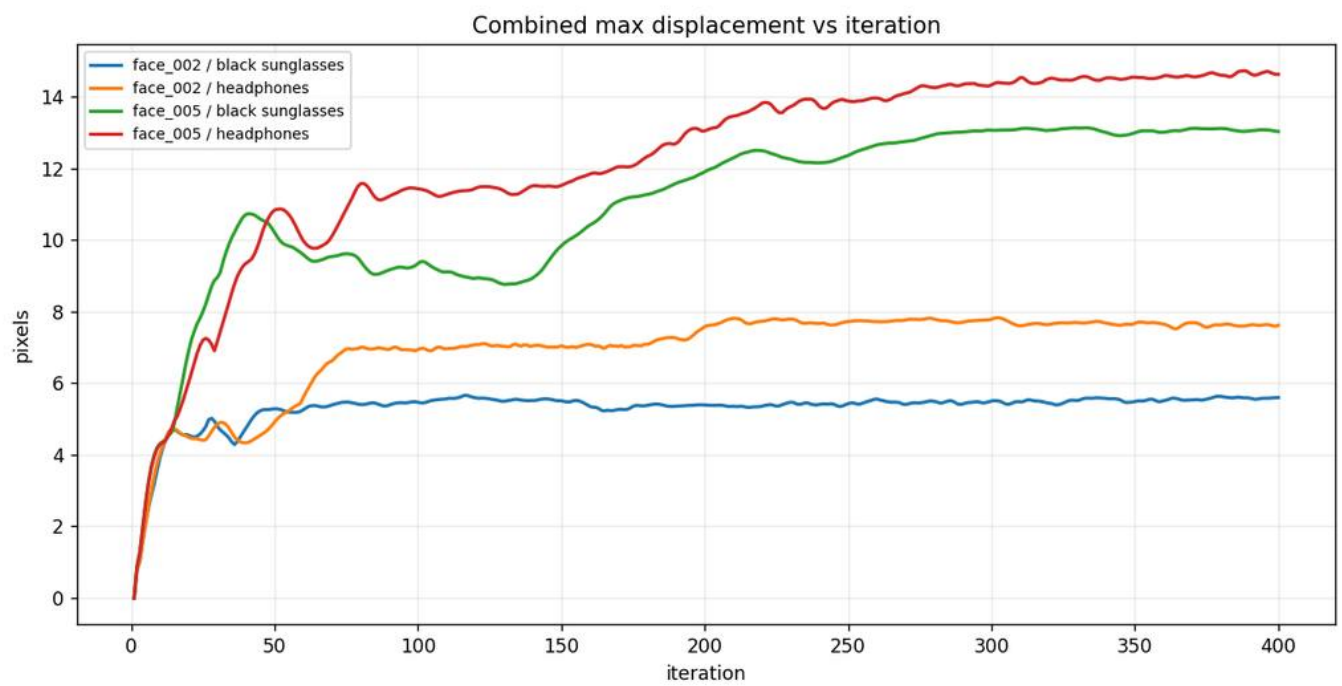
### ***PSNR to original vs iteration***



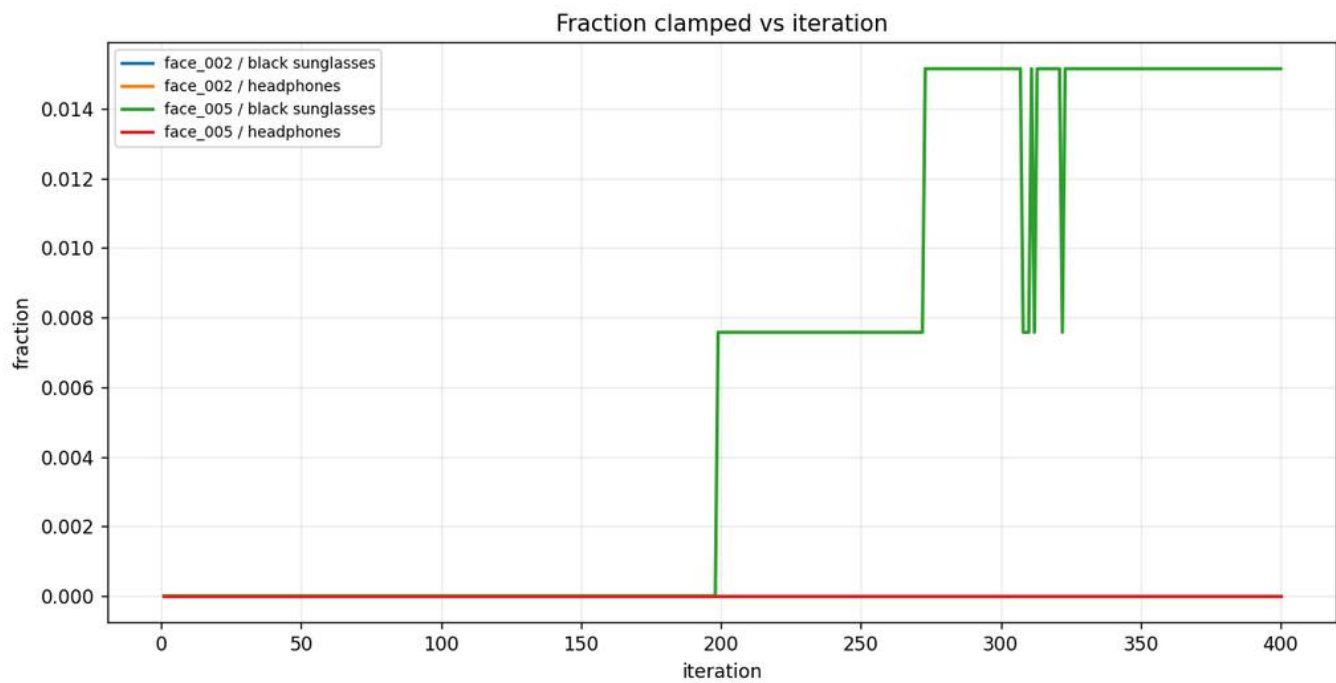
### ***SSIM to original vs iteration***



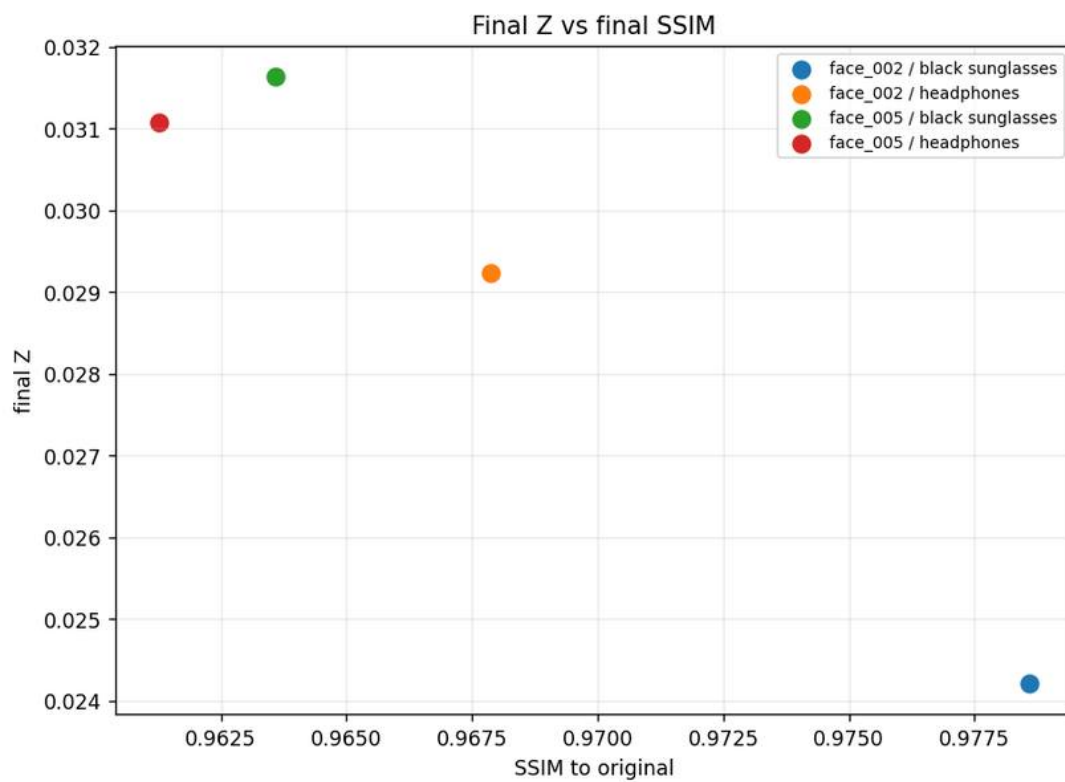
### ***Combined max displacement vs iteration***



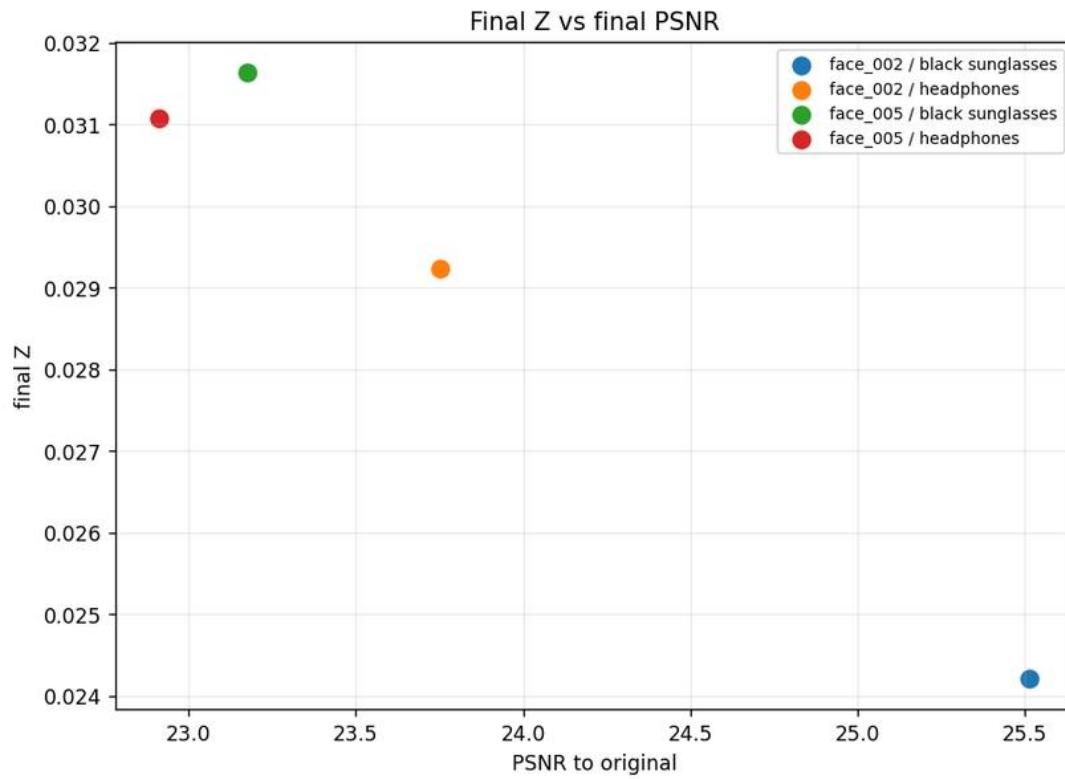
### ***Fraction clamped vs iteration***



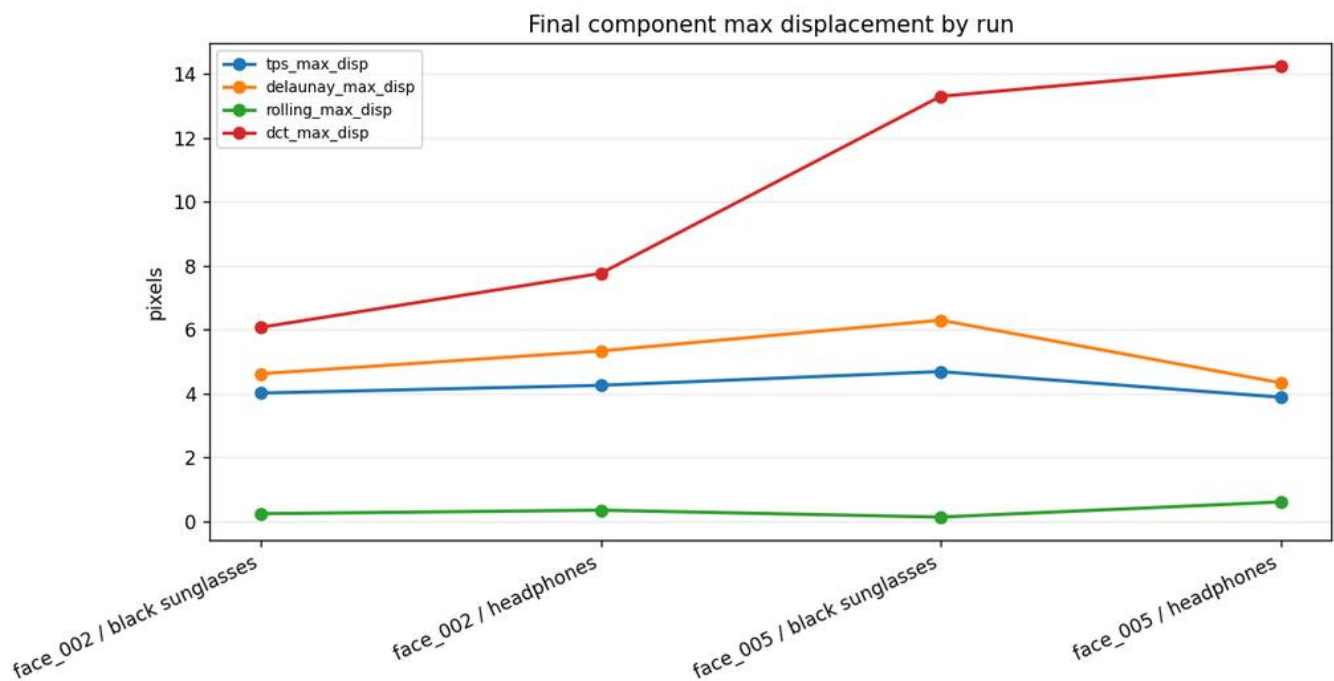
### *Final Z vs final SSIM*



### *Final Z vs final PSNR*



## Component max displacement



## Notes

Completed runs collected: 4.

Run root: outputs/arcface\_identity/20260703\_100609\_arcface\_identity\_all\_sequential



Missing artifacts recorded: 0.